

Product Description

ATEK656N5 is a sub-octave switchable filter bank covering 2 - 18 GHz frequency band. Filterbank consists of an SP6T switch followed by 6 fixed frequency band pass filters which are followed by another SP6T switch.

RF Input and Output are internally matched to 50 ohms for ease of use.

Filterbank provides 33 dBm IIP3 which allows users to realize high dynamic range wideband receiver frontends. Sub-octave filterbank architecture improves overall system IIP2.

Filterbank is housed in 5x5 mm low-cost surface mount package.

Evaluation Board, custom package and module options are available upon request.

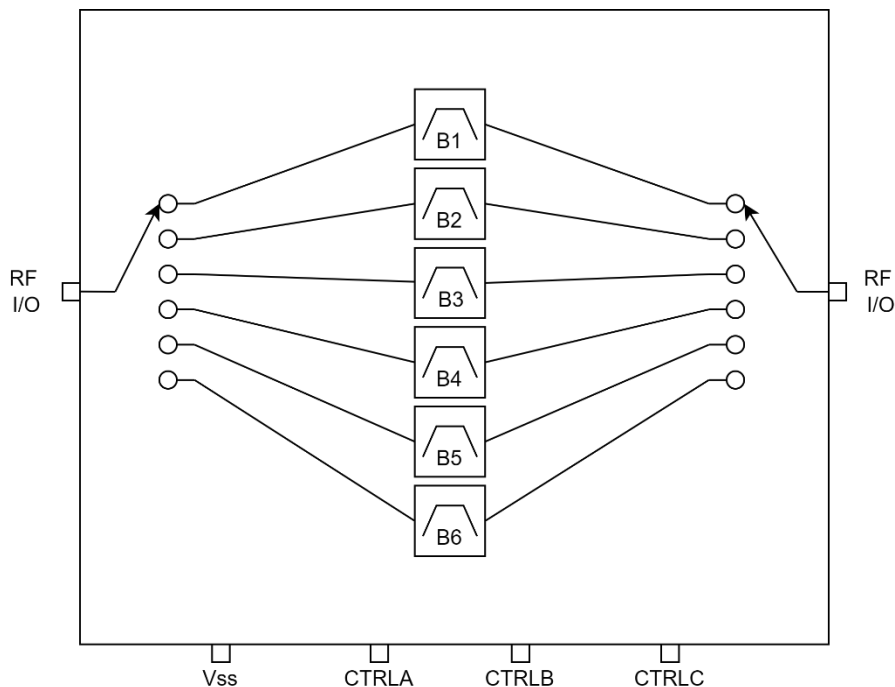
Product Features

- Frequency Range: 2 - 18 GHz
- 6 Switchable Band Pass Filters
- Input IP3: 33 dBm
- 5x5 mm compact size

Applications

- Wideband Receivers
- Telecommunication
- Test and Measurement
- SATCOM
- SDR

Functional Block Diagram



Electrical Specifications

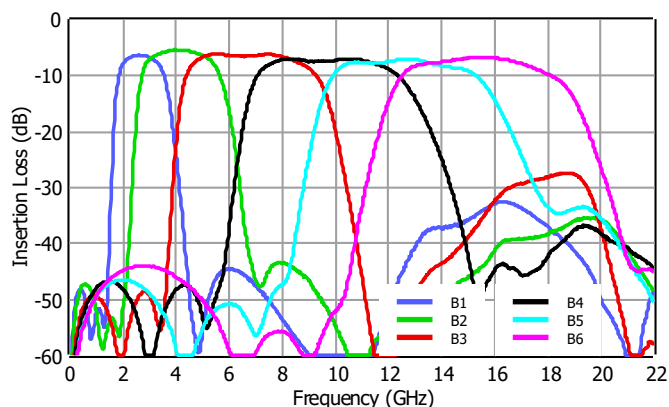
Conditions unless otherwise specified: $V_{SS} = -5\text{ V}$, Typical, $T = 25\text{ }^{\circ}\text{C}$, CW.

Parameter		Min	Typ	Max	Units
Operational Frequency Range		2		18	GHz
3dB Bandwidth (High Pass 3dB cutoff - Low Pass 3dB cutoff)	Band 1		1.9 - 3.5		GHz
	Band 2		2.8 - 5.4		
	Band 3		4.5 - 9.1		
	Band 4		7.1 - 12.3		
	Band 5		9.9 - 15.3		
	Band 6		12.5 - 18		
Insertion Loss	Band 1		7.5		dB
	Band 2		6.5		
	Band 3		6.5		
	Band 4		7.5		
	Band 5		9		
	Band 6		9		
Input Return Loss			-12		dB
Output Return Loss			-10		dB
Input IP3			33		dBm
Input P1dB			23		dBm
Switching Speed	On		120		ns
50% Vctrl to 90% of RF Output	Off		25		
DC Supply Voltage (V_{SS})			-5		V
DC Supply Current			12		mA
Control Voltage (CTRL)	Low		0		V
	High		-5		
Operating Temperature		-40		85	$^{\circ}\text{C}$

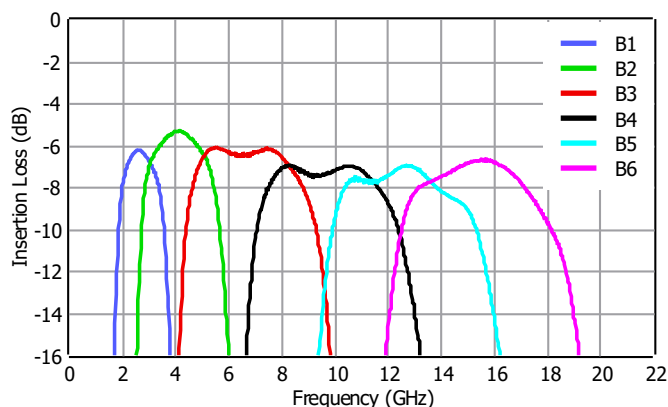
Typical Performance Plots

Conditions unless otherwise specified: $V_{SS} = -5V$, Typical, $T = 25^\circ C$, CW.

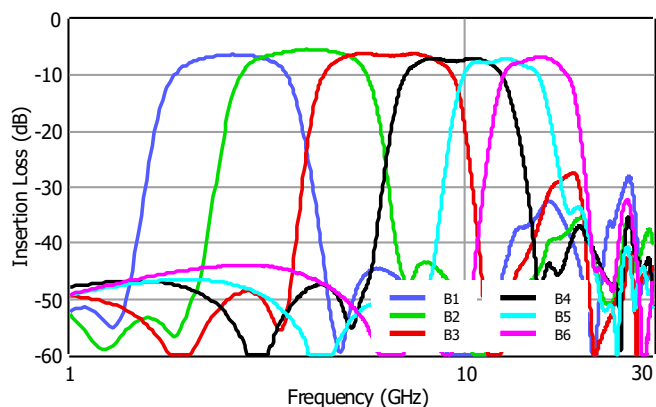
Insertion Loss



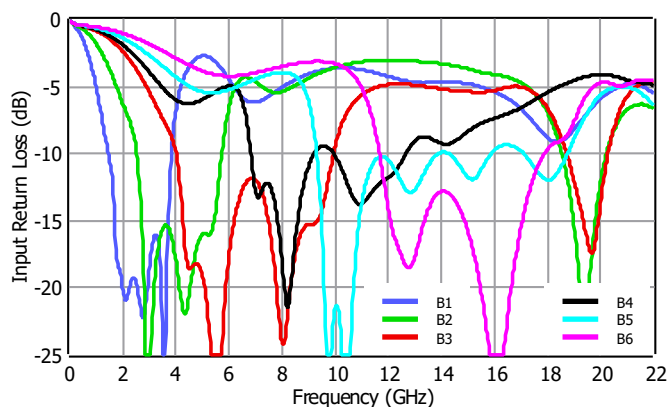
Insertion Loss, Zoom



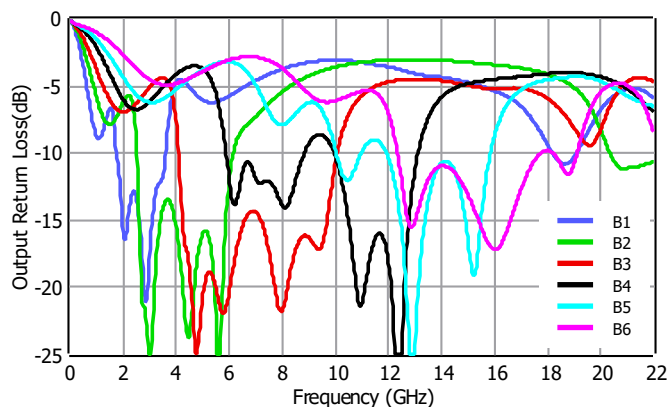
Insertion Loss, Log Scale



Input Return Loss



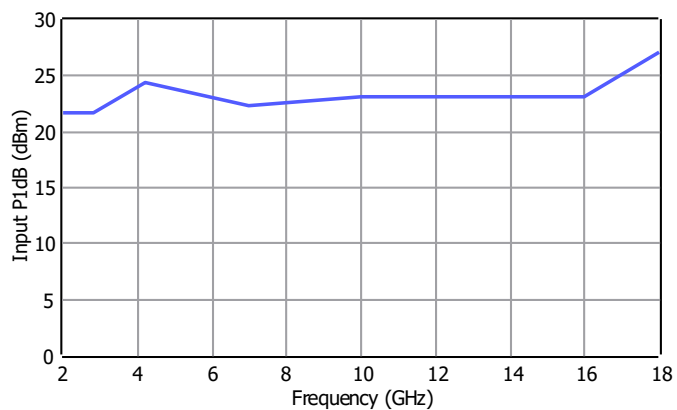
Output Return Loss



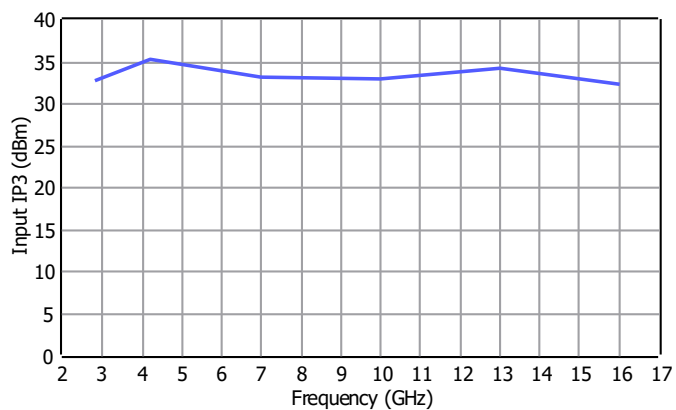
Typical Performance Plots

Conditions unless otherwise specified: $V_{SS} = -5\text{ V}$, Typical, $T = 25\text{ C}$, CW.

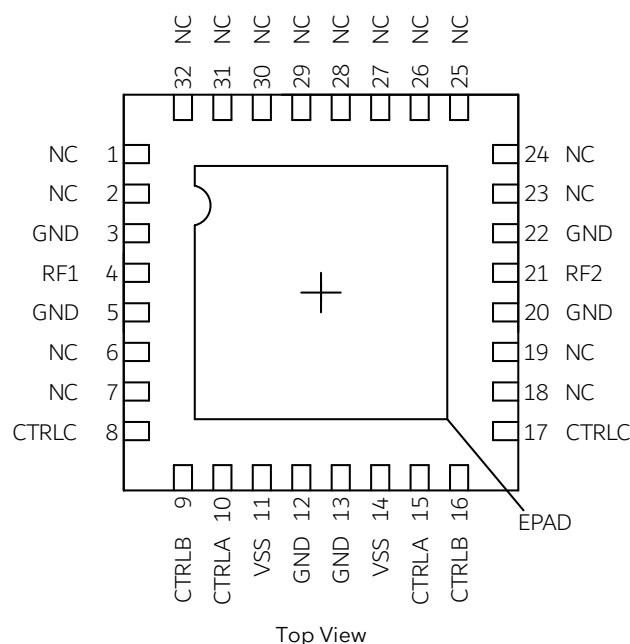
Input P1dB



Input IP3



Pin Description



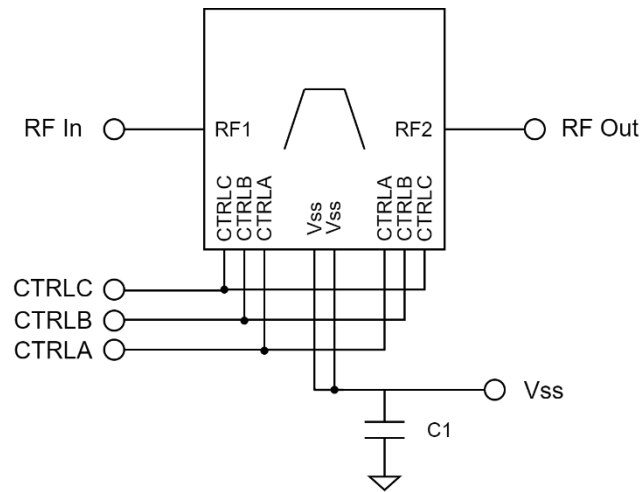
Pin Number	Pin Name	Description
4	RF1	RF input/output pin. If the DC voltage level on RF lines is not equal to 0 V, an external DC block capacitor is required.
21	RF2	RF input/output pin. If the DC voltage level on RF lines is not equal to 0 V, an external DC block capacitor is required.
11, 14	VSS	Voltage supply pin.
10, 15	CTRLA	Control pin. Two CTRLA pins are shorted internally. Using one of them is sufficient for controlling filter state.
9, 16	CTRLB	Control pin. Two CTRLB pins are shorted internally. Using one of them is sufficient for controlling filter state.
8, 17	CTRLC	Control pin. Two CTRLC pins are shorted internally. Using one of them is sufficient for controlling filter state.
1, 2, 6, 7, 18, 19, 23-32	NC	These pins are not internally connected. Can be grounded on the PCB.
3, 5, 12, 13, 20, 22	GND	Ground.
33	EPAD	Exposed Pad on the bottom of the package should be connected to ground with multiple number of vias to reduce the inductance to the GND.

Control Interface

CTRLA	CTRLB	CTRLC	Filter Bank State
HIGH	HIGH	HIGH	Band1
LOW	HIGH	LOW	Band2
HIGH	LOW	HIGH	Band3
LOW	LOW	HIGH	Band4
LOW	HIGH	HIGH	Band5
HIGH	HIGH	LOW	Band6

Applications Information

Signal entering from RF1 pins goes into 6 selectable paths. All these paths are fixed frequency band pass filters. Typical application schematic to operate the filterbank is given below.



C1 is used to filter out the ripples and unwanted signals coming from the Vss supply. Using additional capacitors in parallel to C1 will improve this filtering. If this filtering is of no concern, then filterbank can be operated without C1.

Filterbank can be supplied in alternative packages and custom housing.

CTRL voltages are used for setting the switch state.

Small signal plots are gathered with probe PCB measurements, to generate data shown in this document, where RF1 is input and RF2 is the output.

Large signal data is generated with connectorized evaluation PCB measurements. Then the PCB trace and connector transition losses are de-embedded, to generate plots shown in this document.

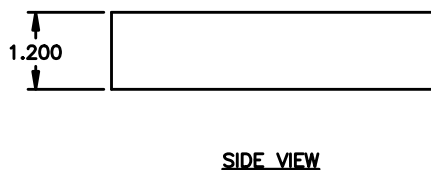
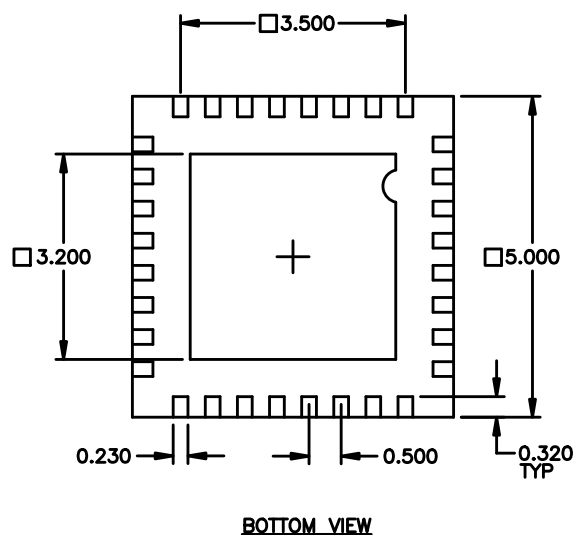
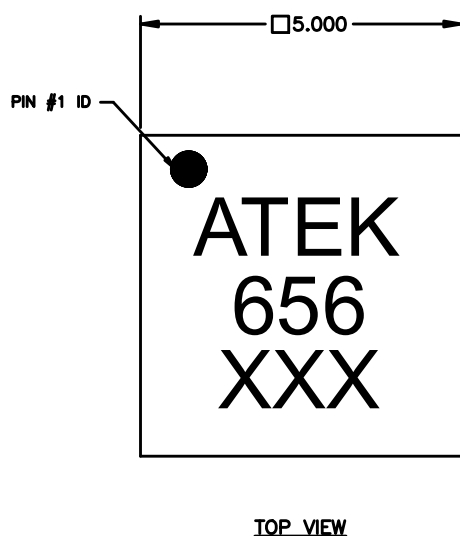
The NC pins of the Filterbank are connected to the GND on the PCBs used to generate the plots shown in this document.

Absolute Maximum Ratings

Parameter	Value/Range
Supply Voltage (Vdd)	TBD
RF Input Power	TBD
Storage Temperature	-55 to +125°C

Operation of this device outside the parameter ranges given above may cause damage. These conditions should not be applied simultaneously.

Mechanical and Marking Information



NOTES

1. ALL DIMENSIONS IN MM

Handling Precautions



Caution!
ESD-Sensitive Device
Handle Accordingly

Contact Information

For the latest specifications, additional product information, support, and sales.

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Revisions

Revision No	Revision Date	Revision Reason	Section / Page No
1.0	05.07.2024	Initial Release	
1.1	05.08.2024	Plots and Electrical Specifications Sections Updated, Pin Description and Applications Information Sections Added	
1.2	30.01.2025	Switching Speed Data Added	
1.3	14.01.2026	Mechanical and Marking Information Revised	
1.4	04.02.2026	Pin Description Updated	